

POWER ELECTRONICS

Controlled Rectifiers

Introduction

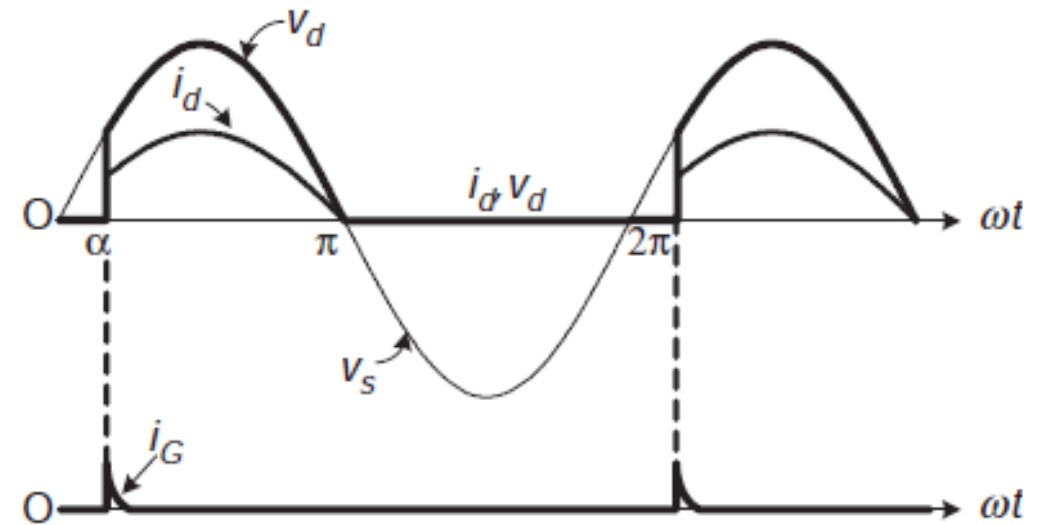
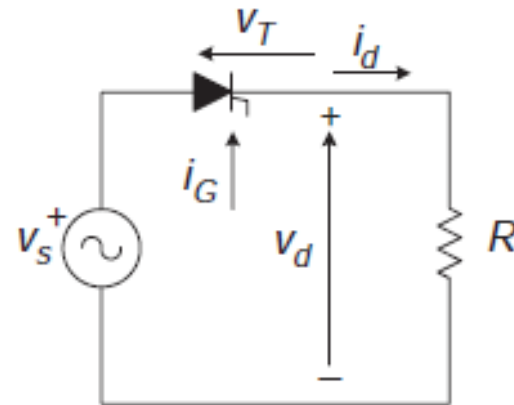
- Diode rectifiers do not allow the control of power being delivered to the load
- Thyristors provide such control

Single-Phase Controlled Rectifiers

- Single-Phase Half-Wave controlled rectifier
- Single-Phase Full-Wave controlled rectifier

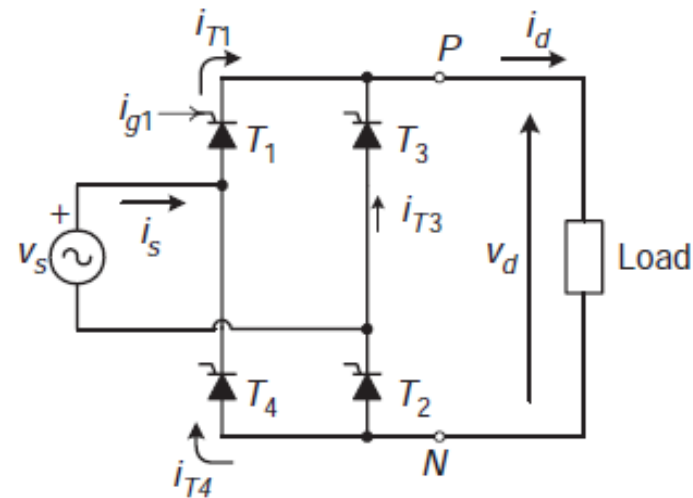
Single-Phase Half-Wave Rectifier

- Firing angle α
- Average/dc value of the load voltage?
- RMS value of the load voltage?

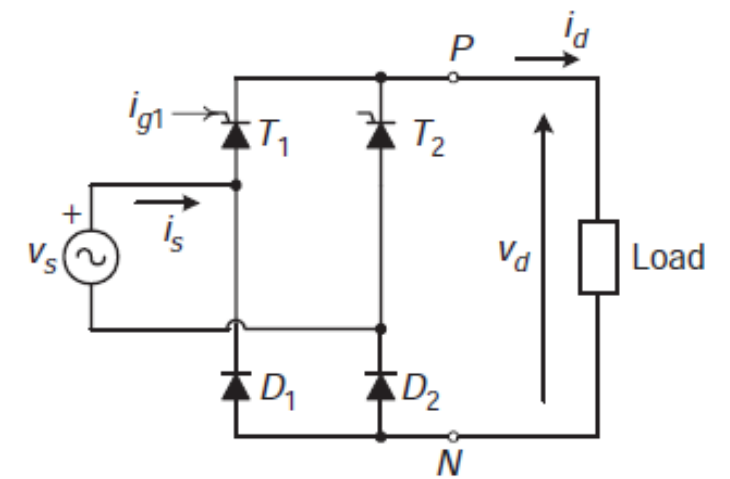


Single-Phase Bridge Rectifier

- Fully controlled
- Half controlled



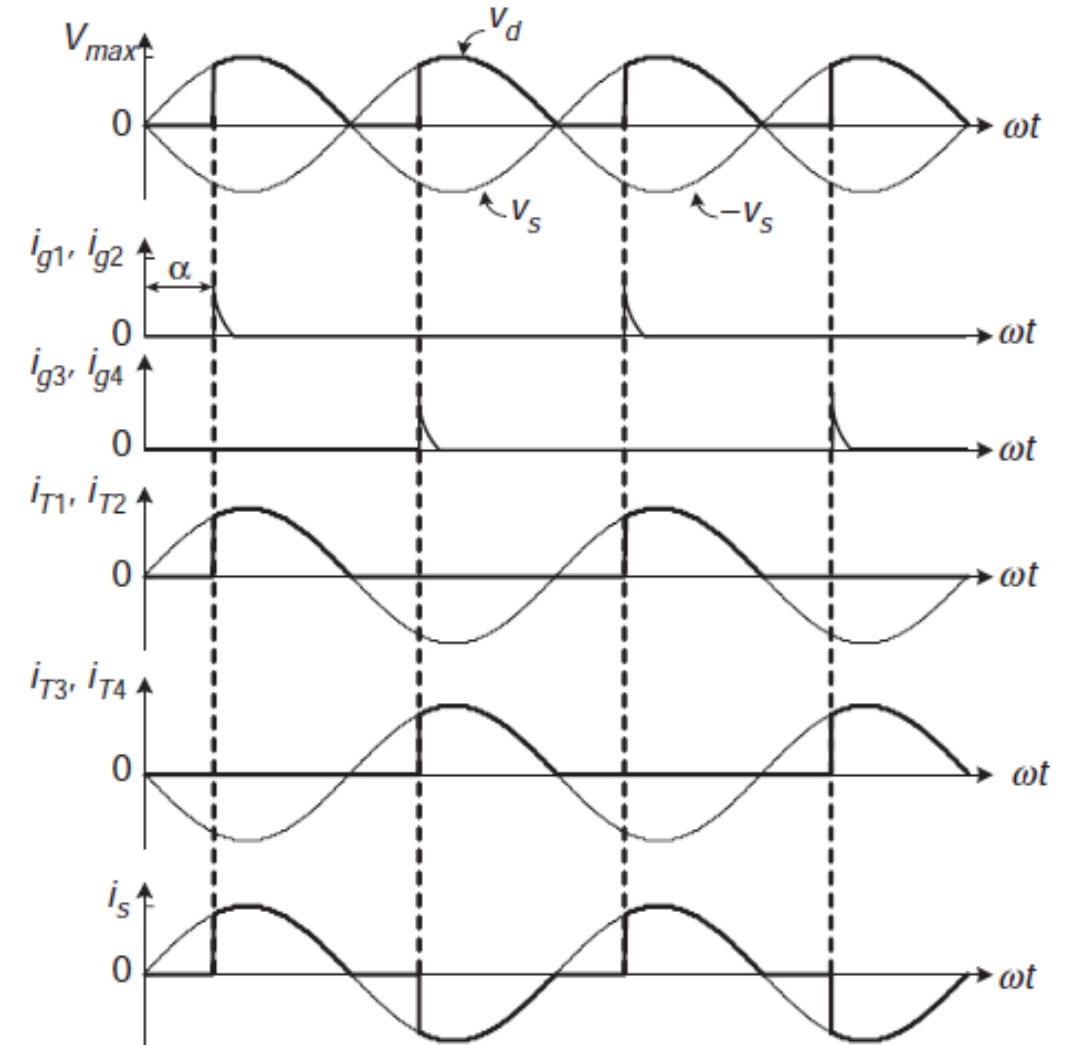
(a)



(b)

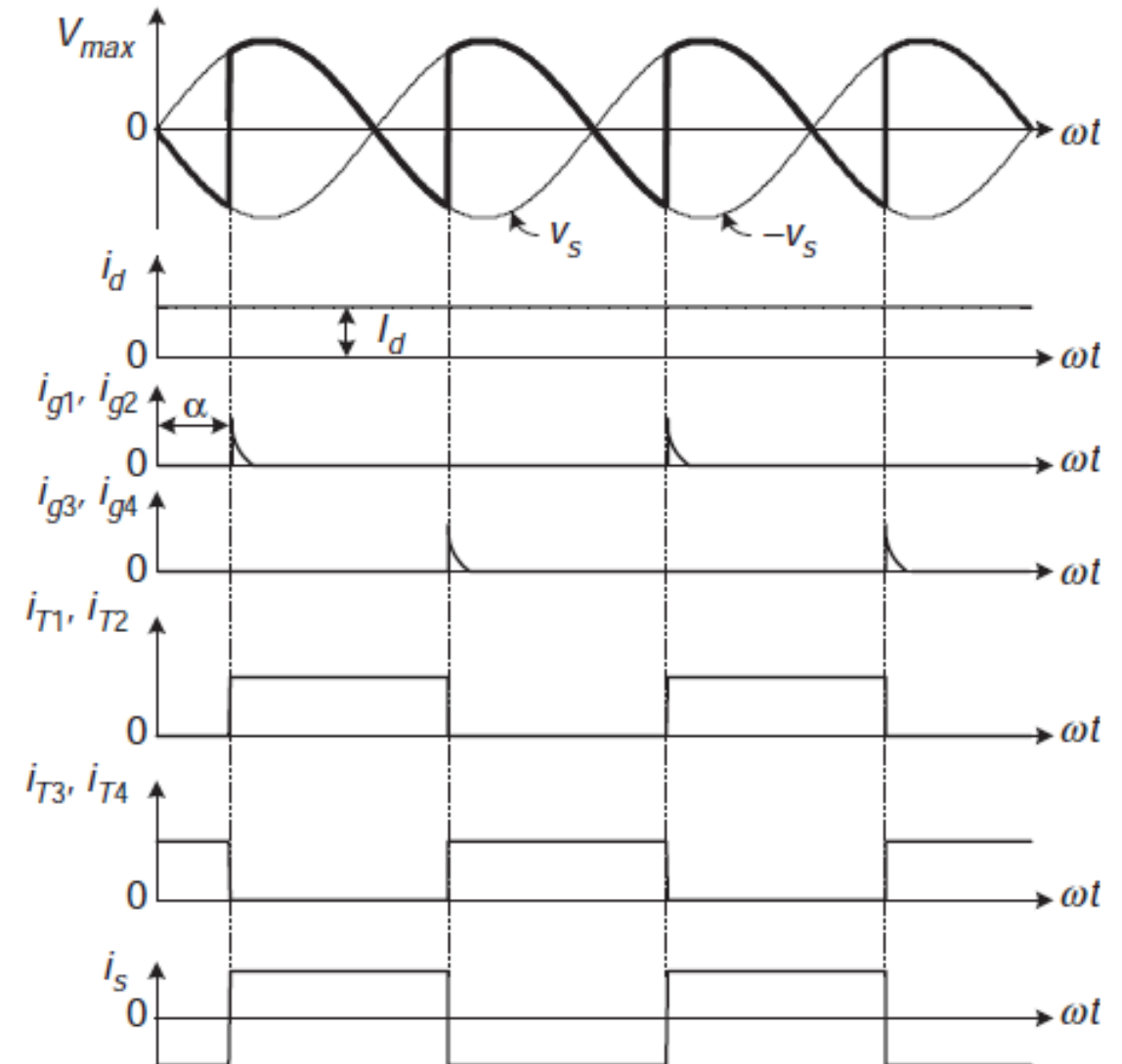
Single-Phase Bridge Rectifier

- Resistive Load
- Fully controlled rectifier waveforms
- Plot the current waveform
- Average output value?
- RMS output value?



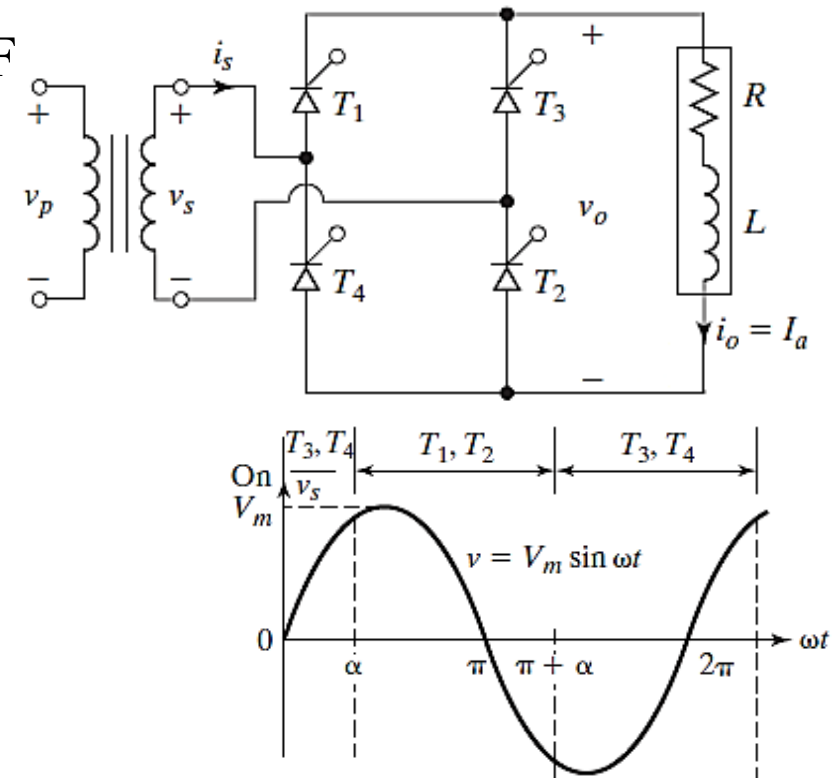
Single-Phase Bridge Rectifier

- RL load ($L \rightarrow \infty$)
- Fully controlled rectifier waveforms
- Average output voltage?
- RMS output voltage?



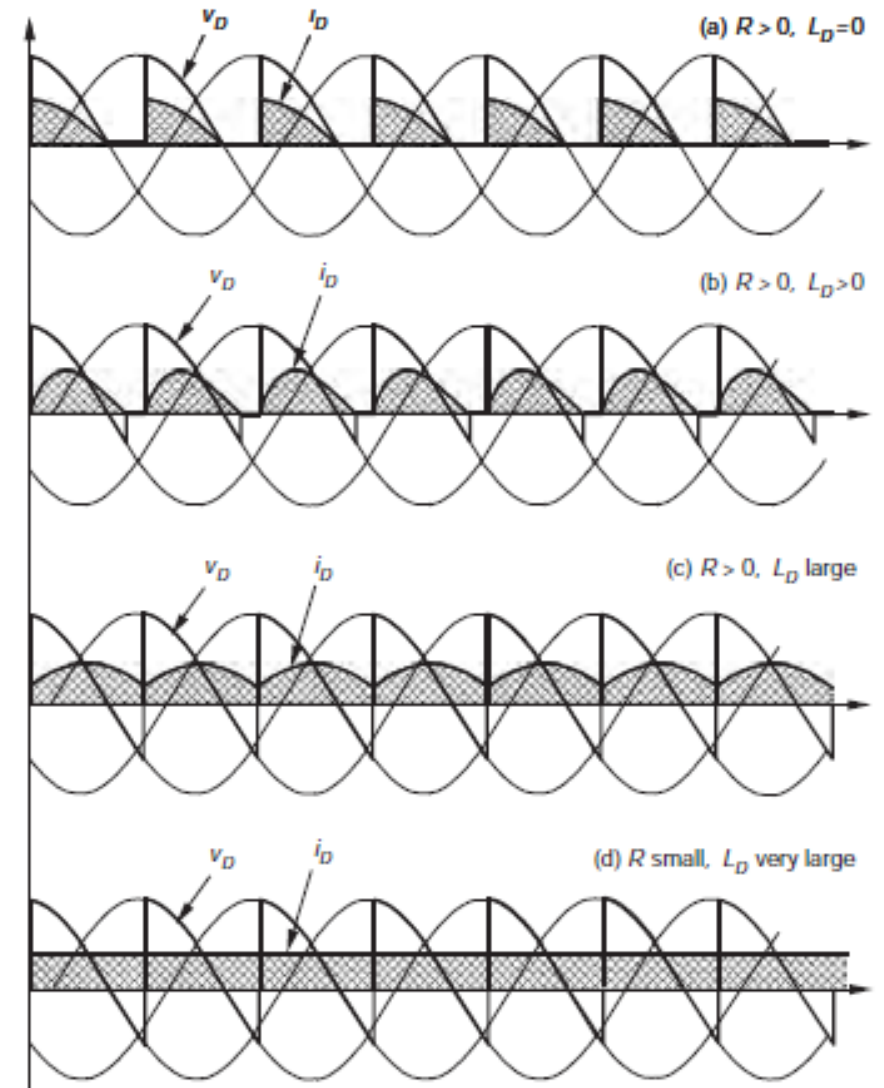
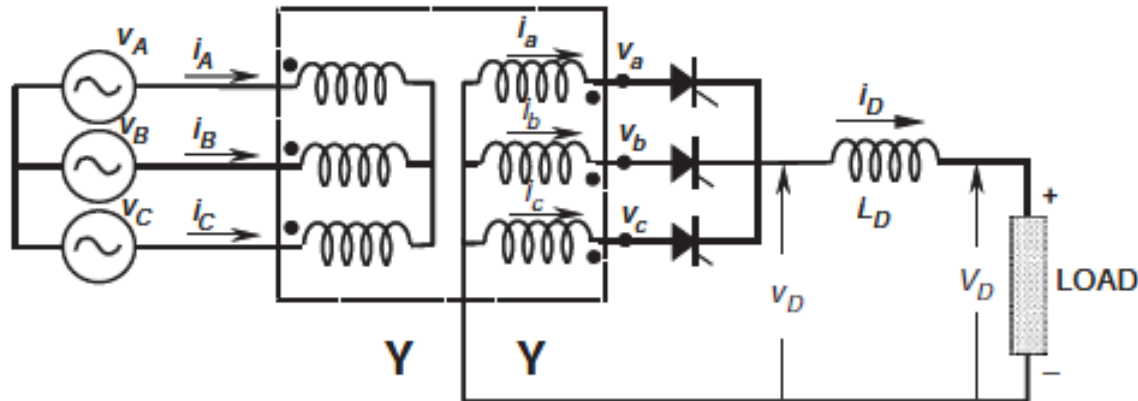
Example

- The full converter shown below is connected to a 120-V, 60-Hz supply. The load current I_a is continuous and its ripple content is negligible. The turns ratio of the transformer is unity. Determine the HF of the input current, DF, and input PF. The fundamental component of the input current is $\frac{2\sqrt{2}I_a}{\pi}$
- If the firing angle is $\alpha = \pi/3$, calculate V_{dc} , V_{rms} , HF, DF and PF

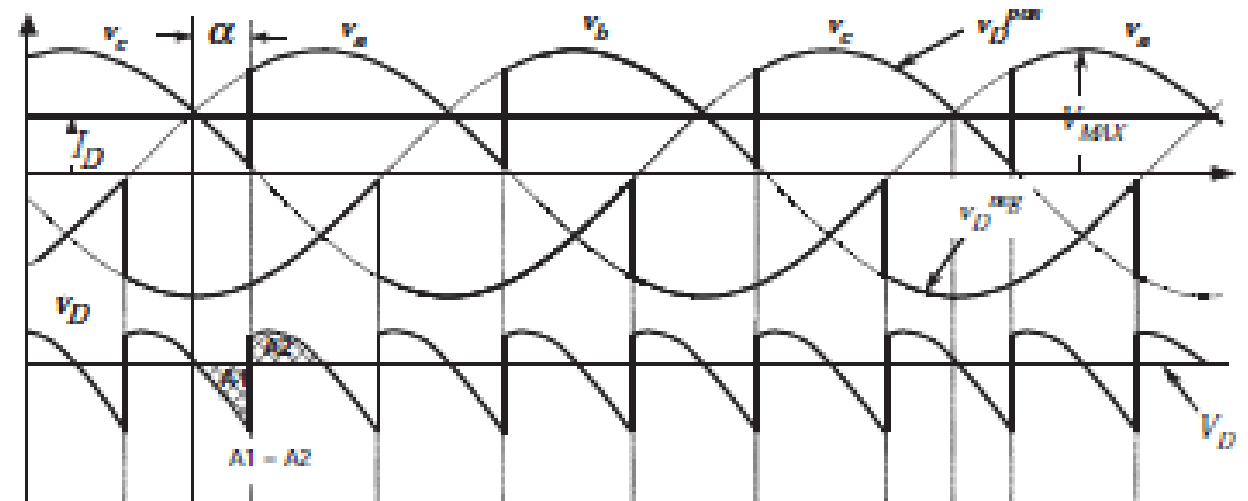


Three-Phase Half-Wave Rectifier

- One SCR per phase



- Two SCRs per phase



Three-Phase Bridge Rectifier

- Average and RMS values?

If the line-to-neutral voltages are defined as

$$v_{an} = V_m \sin \omega t$$

$$v_{bn} = V_m \sin \left(\omega t - \frac{2\pi}{3} \right)$$

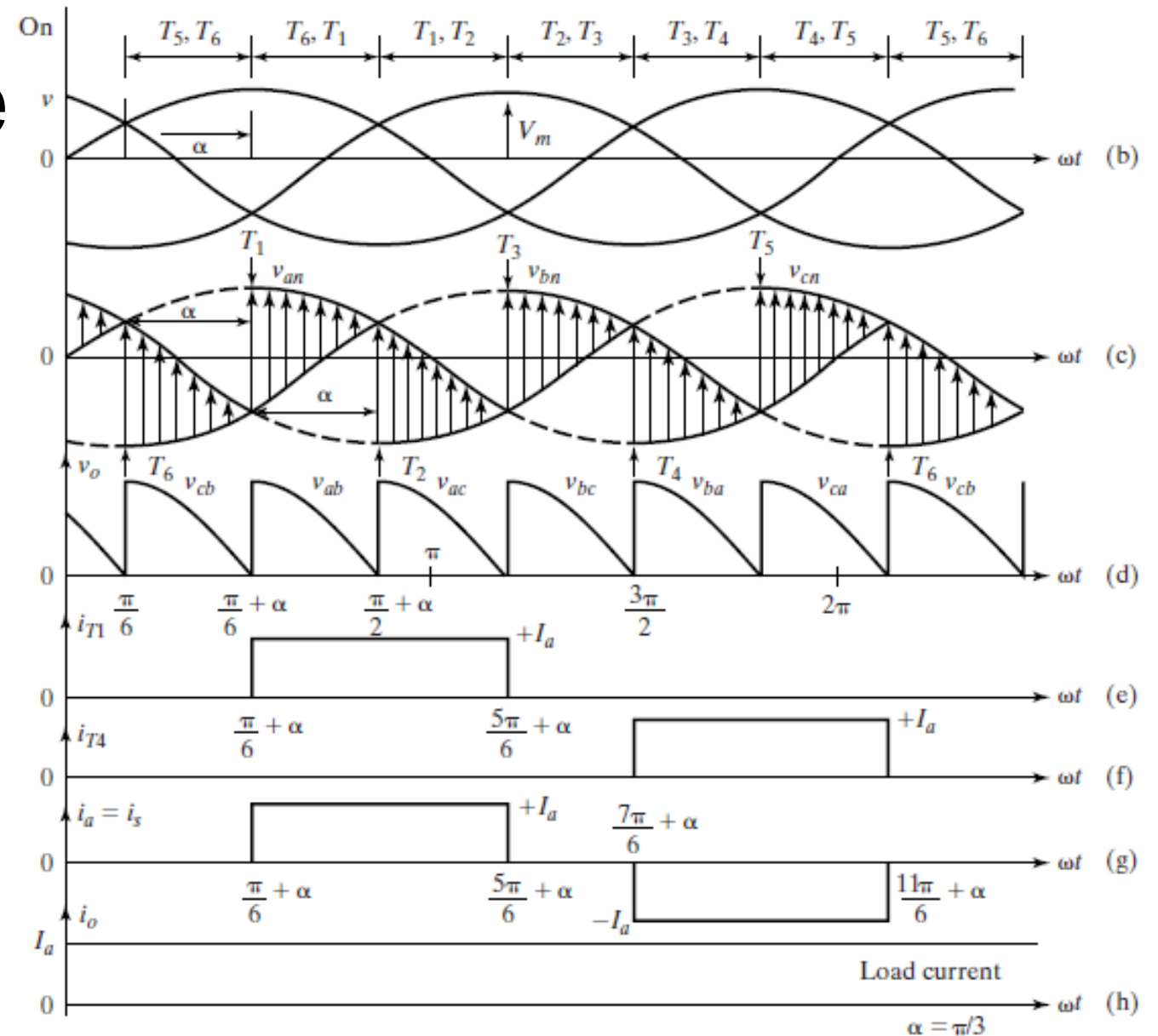
$$v_{cn} = V_m \sin \left(\omega t + \frac{2\pi}{3} \right)$$

the corresponding line-to-line voltages are

$$v_{ab} = v_{an} - v_{bn} = \sqrt{3} V_m \sin \left(\omega t + \frac{\pi}{6} \right)$$

$$v_{bc} = v_{bn} - v_{cn} = \sqrt{3} V_m \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$v_{ca} = v_{cn} - v_{an} = \sqrt{3} V_m \sin \left(\omega t + \frac{\pi}{2} \right)$$



Three-Phase Bridge Rectifier

- Average and RMS values?

Example

- A three-phase full-wave converter is operated from a three-phase Y-connected 208 V, 60 Hz supply and the load resistance is $R = 10 \Omega$. If it is required to obtain an average output voltage of 50% of the maximum possible output voltage, calculate
 - a. The delay angle α
 - b. The rms and average output currents
 - c. The average and rms thyristor currents
 - d. The rectification efficiency
 - e. The TUF
 - f. The input PF

Reference

Power Electronics: Devices, Circuits, and Applications
Haroon Rashid

Power Electronics Handbook
Haroon Rashid